8.2 Stem-and-Leaf Plots

Name _____ Period ____ Date _____

Algebra I 8.2 · Student sheet

I can…

- make a stem-and-leaf plot.
- use a stem-and-leaf plot to find measures of center and spread.
- describe the shape of a distribution as symmetric, skewed left, or skewed right.

Vocabulary · fill in the missing word

Stem-and-leaf plot — a display that uses the digits of the data values to _____ them; the leaves are in order

Stem — every digit _____ the last (for 2-digit data, the tens digit)

Leaf — the last _____ (the ones digit)

Key — a _____ that shows what a stem and leaf mean, such as 3 | 4 = 34

Symmetric — the left and right sides of the _____ are about mirror images

Skewed right — most data on the left, with a tail _____ to the right (larger values)

Skewed left — most data on the right, with a tail _____ to the left (smaller values)

Part 1 · Sort by tens

Minutes 15 students read last night: 23, 35, 41, 18, 27, 32, 45, 29, 38, 26, 33, 21, 40, 37, 30

Group the values by their tens digit. Write each row in order.

Stem	Leaves
1	
2	
3	
4	

Key: 1 | 8 = _____ How many read 30 minutes or more? _____ Median: _____ Range: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Stems and leaves: The **stem** is every digit except the _____; the **leaf** is the _____ digit.

Key: Every plot needs a _____, such as 3 | 4 = 34.

Example 1

Make a stem-and-leaf plot of the points a player scored in 12 games:

12, 25, 18, 31, 20, 14, 27, 22, 19, 33, 25, 16

Example 2

The plot shows daily high temperatures (°F).

a. On how many days was the high above 65°F?

b. Find the mean, median, mode, and range.

Daily High Temperatures (°F)

Stem	Leaf
5	2 5 8
6	0 1 4 4 7 9
7	0 3 5
8	1

Key: 5 | 2 = 52°F

Skewed right: the tail stretches toward the _____ values. **Skewed left:** the tail stretches toward the _____ values.

Example 3

Describe the shape of the distribution.

Stem	Leaf
1	0 1 1 2 3 4 5 6
2	0 2 4 7
3	1 5
4	8

Key: 1 | 0 = 10

Mean in a skewed distribution: In a skewed distribution, the mean is pulled toward the _____.

Example 4

Use the plot from Example 3. Find the mean and the median. Which is greater? Why?

Stem	Leaf
1	0 1 1 2 3 4 5 6
2	0 2 4 7
3	1 5
4	8

Key: 1 | 0 = 10

Part 3 · Partner practice

Partner 1

Make a stem-and-leaf plot: 45, 52, 38, 47, 51, 39, 44, 56, 48, 50

Partner 2

Find the mean, median, mode, and range of the data.

Stem	Leaf
2	4 6 8
3	0 3 3 5 9
4	1 2

Key: 2 | 4 = 24